Exercises 4.3 1, 9, 13, 15, 17, 21, 37(a)(b)(c), 43(a)(b)(c), 45(a)(b)(c), 47(a)(b)(c), 75, 83.

1–2 Use the given graph of f to find the following.

(a) The open intervals on which f is increasing.

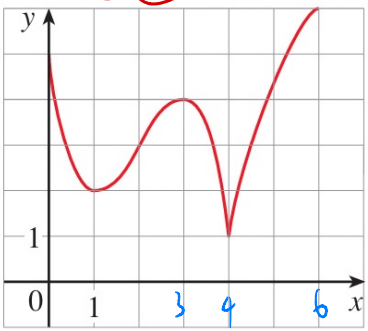
(b) The open intervals on which f is decreasing.

(c) The open intervals on which f is concave upward.

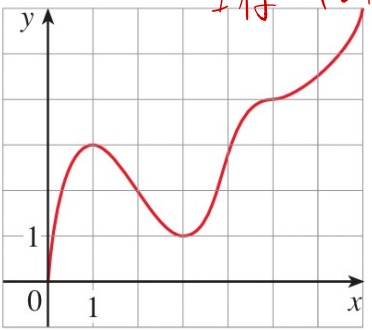
(d) The open intervals on which f is concave downward.

(e) The coordinates of the points of inflection.

1.



2.



9–16 Find the intervals on which f is increasing or decreasing, and find the local maximum and minimum values of f.

9. $f(x) = 2x^3 - 15x^2 + 24x - 5$

$f'(x) = 6x^2 - 30x + 24 = 0$

$6(x^2 - 5x + 4) = 0$

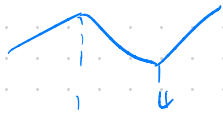
$6(x-4)(x-1) = 0$

$x = 4$ or $x = 1$



when $x < 1$ or $x > 4$, $f'(x) > 0$, $f(x)$ ↑

when $1 < x < 4$, $f'(x) < 0$, $f(x)$ ↓



when $x = 1$, there is a local maximum $f(1) = 6$

$x = 4$ minimum $f(4) = -21$

9–16 Find the intervals on which f is increasing or decreasing, and find the local maximum and minimum values of f.

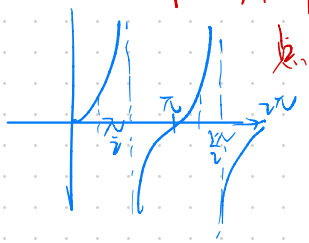
15. $f(x) = \sin x + \cos x, \quad 0 \leq x \leq 2\pi$ 划分为 $(0, \frac{\pi}{4})$, $(\frac{\pi}{4}, \frac{5\pi}{4})$, $(\frac{5\pi}{4}, 2\pi)$, 分区间取点判断

$f'(x) = \cos x - \sin x = 0$

$\cos x = \sin x$

$\tan x = 1$

$x = \frac{\pi}{4}$ 或 $x = \frac{5\pi}{4}$



when $0 < x < \frac{\pi}{4}$ or $\frac{\pi}{2} < x < \frac{5\pi}{4}$ or $\frac{3\pi}{2} < x < 2\pi$, $f'(x) < 0$, $f(x) \downarrow$

$\frac{\pi}{4} < x < \frac{\pi}{2}$ or $\frac{5\pi}{4} < x < \frac{3\pi}{2}$, $f'(x) > 0$, $f(x) \uparrow$ ×

when $x = \frac{\pi}{4}, \frac{5\pi}{4}$, $f(x)$ has local minimum value

$f(\frac{\pi}{4}) = \sin\frac{\pi}{4} + \cos\frac{\pi}{4} = \sqrt{2}$

$f(\frac{5\pi}{4}) = \sin(\frac{5\pi}{4}) + \cos(\frac{5\pi}{4}) = -\sqrt{2}$

不能用 $\tan x > 1$ 判断增减. ∵ $\cos x - \sin x$ 在 $(0, 2\pi)$ 上处处有定义且连续的.

$x = \frac{\pi}{2}, \frac{3\pi}{2}$, $f(x)$ — maximum

但 $\tan x$ 在 $x = \frac{\pi}{2}$ 和 $x = \frac{3\pi}{2}$ 处无定义

$f(\frac{\pi}{2}) = \sin(\frac{\pi}{2}) + \cos(\frac{\pi}{2}) = \frac{\sqrt{2}+1}{2} = f(\frac{3\pi}{2})$

$(0, \frac{\pi}{4})$, $x = 0$, $f'(0) = \cos(0) - \sin(0) = 1 > 0$, $f(x) \uparrow$

$(\frac{\pi}{4}, \frac{5\pi}{4})$, $x = \pi$, $f'(\pi) = \cos(\pi) - \sin(\pi) = -1 < 0$, $f(x) \downarrow$

$(\frac{5\pi}{4}, 2\pi)$, $x = \frac{3\pi}{2}$, $f'(\frac{3\pi}{2}) = \cos(\frac{3\pi}{2}) - \sin(\frac{3\pi}{2}) = +1 > 0$, $f(x) \uparrow$

∴ $x = \frac{\pi}{4}$, local maximum, $f(\frac{\pi}{4}) = \sqrt{2}$

$x = \frac{5\pi}{4}$, local minimum, $f(\frac{5\pi}{4}) = -\sqrt{2}$

17–22 Find the intervals on which f is concave upward or concave downward, and find the inflection points of f.

17. $f(x) = x^3 - 3x^2 - 9x + 4$

$f'(x) = 3x^2 - 6x - 9$

$f''(x) = 6x - 6 = 0$

$x = 1$

when $x < 1$, $f''(x) < 0$. $f(x)$ concave downward

$x > 1$, $f''(x) > 0$, $f(x)$ concave upward.

$\therefore x = 1$ is an inflection point ✓

21. $f(x) = \ln(x^2 + 5)$

$f'(x) = \frac{1}{x^2+5} (x^2+5)' = \frac{2x}{x^2+5}$

$f''(x) = \frac{-2x}{(x^2+5)^2} = 0$ ✗ $\frac{(x^2+5)2 - 2x \cdot 2x}{(x^2+5)^2} = -\frac{2(x^2-5)}{(x^2+5)^2}$

$x = 0$

$x < 0$, $f''(x) > 0$, $f(x)$ concave upward

> 0 < 0 downward.

$x = 0$ ..

43–44 The graph of the derivative f' of a continuous function f is shown.

(a) On what intervals is f increasing? Decreasing?

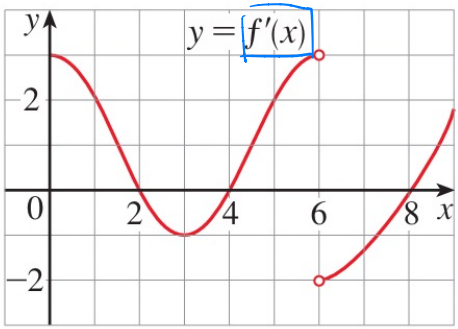
(b) At what values of x does f have a local maximum? Local minimum?

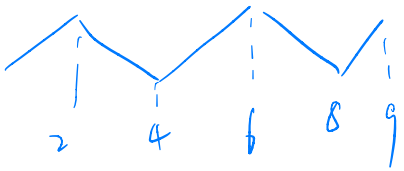
(c) On what intervals is f concave upward? Concave downward?

(d) State the x-coordinate(s) of the point(s) of inflection.

(e) Assuming that $f(0) = 0$, sketch a graph of f.

43.





a) $x \in [0, 2) \cup (4, 6) \cup (8, 9)$ ↑

$x \in [2, 4] \cup (6, 8)$ ↓

b) $x = 2, 6$, $f(x)$ have local maximum

$x = 4, 8$, $f(x)$ — — minimum

c) $x \in [0, 3)$, $f'(x)$ ↓, $f''(x) < 0$, concave ↓

$x \in [3, 6) \cup (6, 9]$, $f'(x)$ ↑, $f''(x) > 0$, concave ↑